

1. Self-Executing Function

```
(function () {  
  // code inside  
})();
```

- This is an **Immediately Invoked Function Expression (IIFE)**.
- It runs automatically as soon as the script loads.
- Keeps variables and functions **private** to avoid conflicts with other scripts on the page.

2. Create Floating Button

```
const openBtn = document.createElement("button");  
openBtn.id = "floatOpenBtn";  
openBtn.textContent = "Chat";  
document.body.appendChild(openBtn);
```

- Dynamically creates a **button** element.
- Gives it an ID `floatOpenBtn`.
- Sets its text to "Chat".
- Appends it to the page's `<body>`.

Purpose: This button will let the user open the chatbot.

3. Create Floating Chat Window

```
const floatBox = document.createElement("div");  
floatBox.id = "floatBox";  
floatBox.innerHTML = `...`;  
document.body.appendChild(floatBox);
```

- Creates a **div** that will be the floating chat window.
- Assigns it an ID `floatBox`.
- Fills it with HTML for:
 - Close button ×
 - Bot title: 💎 Shop Buddy Bot
 - Chat messages area
 - Input field + Send button
- Appends it to the page.

HTML Inside the Floating Box

```
<div id="chatbox">  
  <div id="messages">  
    <div class="msg bot">Hello 🤖! Ask about products...</div>
```

```

</div>
<div id="inputArea">
  <input id="userInput" placeholder="Type your message...">
  <button id="sendBtn">Send</button>
</div>
</div>

```

- #messages: Area where chat messages appear.
- #inputArea: Contains input field + send button.
- First message is from the **bot** as a welcome message.

4. Add Styles

```

const style = document.createElement("style");
style.textContent = `...`;
document.head.appendChild(style);

```

- Adds **CSS dynamically** to the page.
- Styles include:
 - Floating button: round, shadow, pulse animation
 - Chat window: box shadow, border radius, fixed position
 - Messages: different background for bot vs user
 - Animations: fadeIn, slideMsg, pulse

5. Open & Close Functionality

```

openBtn.onclick = () => {
  floatBox.style.display = "block";
  openBtn.style.display = "none";
};

document.addEventListener("click", (e) => {
  if (e.target.id === "floatCloseBtn") {
    floatBox.style.display = "none";
    openBtn.style.display = "block";
  }
});

```

- **Open button click:** Shows the chat window and hides the floating button.
- **Close button click:** Hides the chat window and shows the floating button again.

6. Chatbot Logic

```

setTimeout(() => {
  const HF_TOKEN = "...";

```

```
import("https://cdn.jsdelivr.net/npm/@gradio/client/+esm").then(({ Client })
=> {
    ...
});
}, 500);
```

- Uses `setTimeout` to delay the bot logic by 0.5 seconds (so HTML elements are loaded first).
- Uses **Gradio Client** to connect to Hugging Face chatbot (`clothing_bot`).

Inside Gradio Client

```
const input = document.getElementById("userInput");
const btn = document.getElementById("sendBtn");
const messages = document.getElementById("messages");
```

```
function addMessage(text, sender) {
    const msg = document.createElement("div");
    msg.classList.add("msg", sender);
    msg.textContent = text;
    messages.appendChild(msg);
    messages.scrollTop = messages.scrollHeight;
}
```

- Gets **input, button, and messages div**.
- `addMessage`: Adds a message to the chatbox and scrolls to bottom.
 - `sender` = "user" or "bot" to style differently.

Send Button Click

```
btn.addEventListener("click", async () => {
    const userMessage = input.value.trim();
    if (!userMessage) return;

    addMessage(userMessage, "user");
    input.value = "";
    addMessage("🕒 Thinking...", "bot");

    try {
        const client = await Client.connect("Manish890111/clothing_bot",
        { hf_token: HF_TOKEN });
        const result = await client.predict("/chat", { message:
        userMessage });
        const botMsg = result.data[0] || "Sorry, I didn't understand that.";
        messages.lastChild.textContent = "💬 " + botMsg;
    } catch (err) {
```

```

        messages.lastChild.textContent = "⚠️ Error: " + err.message;
    }
});

```

- **Steps when user clicks Send:**
 - Get input text.
 - Add it to chat as "user".
 - Show "Thinking..." as a bot placeholder.
 - Connect to Hugging Face model (clothing_bot).
 - Send user message to model.
 - Display bot response in chat.
 - Handle errors if connection fails.

Enter Key Press

```

input.addEventListener("keypress", (e) => {
    if (e.key === "Enter") btn.click();
});

```

- Allows sending the message by **pressing Enter** instead of clicking the button.

Summary

- The script **dynamically creates a floating chat button**.
- Opens a **floating chat window** when clicked.
- Styles the button, window, and messages.
- Integrates a **Hugging Face chatbot** using Gradio Client.
- Handles **user messages** and **bot responses** dynamically.
- Works with **click** and **Enter key** input.